

MOHAMED AGHNAOU



Software Engineer seeking a full-time position.

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Software Development Engineer, with a strong background in designing and building end-to-end software solutions. Skilled in backend and frontend development, database design, and application integration. Experienced in developing scalable, maintainable applications and working across the full software lifecycle.

Education

- 📅 2022–2025 Software Development Engineering
National Institute of Posts and Telecommunications (INPT)
- 📅 2020–2022 Preparatory Classes for Engineering Schools (MP)
Lycée Réda Slaoui - Agadir

Professional Experience

ORACLE

📅 01/2025–09/2025 📍 Casablanca, Morocco

Software Engineer – Research Assistant

- Designed and implemented a backend feature enabling users to unlink Intercompany Cross Charge Journals (ICCs) directly from the NetSuite interface.
- Ensured financial data integrity by enforcing accounting period constraints, permission rules, and guaranteeing atomicity of database operations.
- Established a comprehensive testing strategy including unit, component, and E2E tests, ensuring reliability of the feature in a production environment.

Technologies : Java, NetSuite ERP, Git, GitLab, Jira, TeamCity, Oracle Database, SuiteQL, JUnit, Mockito

Royal Air Maroc

📅 06/2024–08/2024 📍 Casablanca, Morocco

Full-Stack Engineer

- Developed a web application promoting sustainable tourism and natural travel destinations.
- Implemented interactive eco-travel exploration and booking features based on dynamic itineraries.
- Designed secure user authentication with real-time reviews.
- Integrated secure online payment processing for reservations.

Technologies : NodeJs, Express, MongoDB, Mongoose, Pug, Git, Github

Projects

Enterprise Knowledge Management AI Assistant

08/2025–10/2025

Design and development of a full-stack AI assistant enabling natural language querying of 500+ internal documents:

- RAG pipeline (chunking, vectorization, reranking) with Spring AI and pgvector, reducing search time by 70%
- AI Agent dynamically orchestrating 4+ tools via MCP (filesystem, external APIs, database)
- Real-time document Q&A interface with SSE streaming, conversation history, and JWT-secured REST API

Technologies: Java, Spring Boot, Spring AI, LangChain4j, OpenAI API, MCP, RAG, Angular, Python, Node.js, pgvector

BiblioManager Application

03/2024–05/2024

Developed an application simplifying book and reservation management:

- Implemented an intuitive system enabling users to add, search, and track available books.
- Developed reservation management and user review features.

Technologies : ReactJs, Bootstrap, Spring Boot, Spring Security, Spring Data JPA, Spring JWT, Okta

ReserveINPT

06/2023–08/2023

Developed a mobile reservation application for managing institutional facilities:

- Enabled users to create, modify, and cancel reservations.
- Provided administrative tools for managing student accounts and facilities.

Technologies : Java, AndroidStudio

Skills

- Programming :** Java, JavaScript, TypeScript, Python, Kotlin
- Backend Development :** JEE, Spring, Spring Boot, Spring Data, Spring Cloud, Spring Security, Keycloak, NodeJs, Express
- Frontend Development :** HTML, Bootstrap, Tailwind CSS, ReactJs, Angular, Next.js
- Generative AI & LLMs :** LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Multimodal AI, OpenAI API, LLaMA
- AI Agents & Automation :** Agentic AI, Model Context Protocol (MCP), Spring AI, Workflow Orchestration, Tool Calling
- DevOps :** Docker, Kubernetes, Kafka, GitLab CI/CD, Jenkins
- Databases :** MySQL, SQL Server, MongoDB, Oracle Database, PostgreSQL
- Software Architectures :** Microservices, Event Driven Architecture, Event Sourcing, CQRS, Saga Pattern OAuth2.0, REST APIs, GraphQL,DDD

Languages

Arabic : Native English : Professional French : Fluent (DELF B2)

Soft Skills

Communication - Teamwork - Problem Solving - Adaptability - Self-learning